

Tech Briefing

ZigZag

Theodor Holm Nelson, Keio University, Southampton University and Project Xanadu

DEEPER COSMOLOGY, DEEPER DOCUMENTS

Computers today basically simulate two things: hierarchy and paper. Hierarchy has been carefully put into the structure of computer files because those who did so considered it right and natural and the only way. Paper has been simulated in the structure of computers because it seemed right and natural and the only way. I believe both of these are forms of imprisonment that constrict and warp our work and our thinking.

Adobe Acrobat and the World Wide Web simulate **both** hierarchy and paper— hierarchy on the left, paper on the right. Acrobat does this in a single package; the Web does it via the URL (which follows the domain name with a hierarchical directory path, then ends in a simulation of paper such as an HTML file). And both these formats glorify appearance over the management of content flow, representing, as it were, the triumph of typesetters over authors.

I would like to propose principled alternatives to both hierarchy and paper.

If we are going to find an alternative cosmology to replace hierarchical directories, it had better be just as simple, just as extensible, just as scalable, and very beneficial.

I think that I have found such a structure. It is both humorous and powerful.

In this cosmology, we build everything out of free-floating cells in n-dimensional space, and view them locally from the cursor outward— spreadsheets let loose and unsprung. These cross-viewable multidimensional lists (trademarked as "ZigZag" in some places) provide an enjoyable new paradigm for all computer structures, with interesting new designs and models for input, viewing, interaction, data structures and programming.

These include—

a paradoxical space with surprising new geometrical properties; a default visualization method that works even in tiny windows; a default interface that works even on phone keypads and game handles; a generalization of lists and tables to N dimensions, providing new forms of database, inheritance, file management, simplified graphical programming, and other useful computer concepts; an enactment engine— a spatial interpreter— for new kinds of programming.

Most importantly, multiple dimensions yield surprising simplifications. These structures appear to streamline many aspects of data and programming, for several reasons— everything may be visualized and manipulated uniformly at the lowest level; special cases may always be mapped to the same primitives; the built-in connection primitives provide pathways that render many customary constructs unnecessary; elements may be referenced in many contexts with no need to spawn or maintain redundant copies or structures.

Naturally, this has a number of applications for hypertext and hypermedia.

I will demonstrate—

- Genealogy where all parties are allowed to have a full set of ancestors; unlike the conventional representation of genealogy by "family trees", where women are spontaneously generated;
- Personal organizers whose elements (like names and addresses) may be directly connected whatever else they need to (like appointments);
- Taking note of a birthday as a fluid item, simultaneously hooking up to name and address, calendar and other birthdays— rather than requiring the opening of three different "applications".
- Operating-system mechanisms easily explorable at all levels.

This system also offers its own new and engrossing set of problems. It is easy to connect things in many ways. It is also easy to get in trouble with the selection of ranks and dimensions which do not scale.

Projects based on this model are underway in Japan, Australia, Finland and the U.K. I hope to make it a whole operating system that does away with "applications" altogether.

Now as to documents.

Project Xanadu, widely misunderstood, has always been about electronic documents with— 2-way links; visible flow of content among documents and versions, especially showing the differences among versions and the origins of quotations and excerpts; frictionless re-use of content; a clean copyright solution.

The specifics of the Xanadu model are extremely simple.

First we establish a Content Layer— a level of:

- cleanly quotable content (no internal markup— keep it on the side),
- stable addressability (divorced from absolute locations),
- long-term availability— a serious and credible promise backed up with appropriate mechanisms.

All new content goes into this content layer, not into the virtual documents or versions themselves.

The rest is very straightforward.

- DOCUMENTS (more properly, Virtual Versions) are distributed in virtual form, as lists of acquisition pointers to this content.

- LINKS are between elements in this content layer. Thus a link may be traced

(by its addresses) from any instance of content on one side of the link to content on the other side; and in principle to any instance of that

content on the other side.

- TRANSLUCIDIONS (recognizable identities between contents of documents and versions, showing origins) are recognized at the system level by identities of content addresses.

- VERSIONING is in principle easily managed: since a version is a list of content, any two versions may be compared by the addresses of their contents to find their hereditary differences (not misled by retyping). Version-intensive occupations, like law, government and movies, constantly need to find differences and origins among versions. Many endeavors require the maintenance of conjoined parallel versions for different audiences. Existing tools are totally irrelevant.

- GROUPWARE is clarified by the exact identifiability of each person's contributions, and the easy simultaneous maintenance of alternative conjoined versions for comparison and emphasis.

- COPYRIGHT: knowing that copyright would not go away, we proposed to make content re-usable and mixable. Our clean copyright model is based on the sale of fragments and the right to use them virtually in new contexts. Micropayments are proportional to downloaded contents (royalty x quantity). This checksums and is manifestly fair.

That is the Xanadu model as we have worked on it for decades. Because it is so different from the conventional approaches, no implementations have been successful to date, though the 1988-92 designs (especially by Roger Gregory, Mark Miller and Eric Drexler) were brilliant.

For nearly the last decade I have worked to adapt this model to the Internet. Recently, David Durand and I have come up with a new proposed format on this model that will be intercompatible with one-way HTML links, though in principle creating new islands of bidirectional connection (both links and translucidions).

This format (tentatively called VLIT, Virtual Literary Format) will be presented here. It uses less intrusive forms of XML and XLINK to specify contents to be brought in and marked up. It requires both server scripting and on-board browser scripting within the file, since it must outwit the mechanisms that were designed from an opposite mentality. It also needs special editors and will benefit from browser plug-ins and a deep caching mechanism for purchased content.

This endeavor is considerably helped by the assumption that the issue of cleanly re-usable stabilized content— content which is markup-free, delocalized and promised— may be separated and handled by a number of alternative parallel means.

These ideas may be disconcerting to those who have steeped themselves in today's special effects. I believe it will be very good news for authors, editors, and those concerned with the integrity, continuity and flow of content, and with the usability of copyrighted material.